

Chapter 4: Polynomials and Rational Fractions

National High School of Mathematics

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1. Polynomials

Let K be a field, such that $K = \mathbb{R}$ or $K = \mathbb{C}$.

1.1 Basic Definitions

Polynomials

The set of all sequences $(a_n)_{n \in \mathbb{N}_0}$ of elements in K such that for some $m \in \mathbb{N}_0$, $a_n = 0$ for all $n \geq m$, is denoted by $K^{\mathbb{N}_0}$. The elements of $K^{\mathbb{N}_0}$ are said to be **polynomials**.

- If we denote an element $(a_n)_{n \in \mathbb{N}_0}$ by P , then a_n is called the **coefficient** of P .
- Two polynomials $P = (a_n)$ and $Q = (b_n)$ are said to be equal if and only if $a_i = b_i$ for all $i \in \mathbb{N}_0$.
- A polynomial $P \in K^{\mathbb{N}_0}$ is said to be **constant** if P is of the form $P = (a_0, 0, \dots)$, and is denoted by $P = a_0$.

Let us denote:

- $X = (0, 1, 0, \dots)$
- $X^2 = (0, 0, 1, 0, \dots)$
- $X^n = (0, \dots, 0, 1, 0, \dots)$ (1 appears in the $(n+1)$ -th position, for index n)

Thus, any polynomial $P = (a_0, a_1, a_2, \dots, a_m, 0, \dots)$ can be written as:

$$P = a_0(1, 0, \dots) + a_1(0, 1, 0, \dots) + \dots + a_m(0, \dots, 1, \dots) = \sum_{k \geq 0} a_k X^k$$

The set of polynomials with coefficients in K is denoted $K[X]$.

1.2 Binary Operations on $K[X]$

Let $P = \sum a_k X^k$ and $Q = \sum b_k X^k$.

1. **Addition:** $P + Q = \sum (a_k + b_k) X^k$
2. **Multiplication:** $PQ = \sum c_n X^n$, where $c_n = \sum_{k=0}^n a_k b_{n-k}$.

We can show that $(K[X], +, \cdot)$ is a **commutative ring with unity**.

1.3 Degree and Valuation

Degree of a Polynomial

The **degree** of a non-zero polynomial $P = \sum a_k X^k$ is the largest integer p such that $a_p \neq 0$. It is denoted by $\deg(P)$.

- The degree of the zero polynomial is defined to be $-\infty$, i.e., $\deg(0) = -\infty$.

Examples

1. $\deg(1 + X^2 + X^4 + X^3) = 4$
2. $\deg(2) = 0$
3. $\deg((X - 1)^n - (X + 1)^n) = n - 1$

Properties of Degree

Let $P, Q \in K[X]$ and $\lambda \in K \setminus \{0\}$.

1. **Degree of Sum:** $\deg(P + Q) \leq \max(\deg(P), \deg(Q))$

- **Proof**

Example Let $P(X) = X^2 + 1$ and $Q(X) = -X^2 + X$. $\deg(P + Q) = \deg(X + 1) =$

1. $\max(\deg(P), \deg(Q)) = \max(2, 2) = 2$. So $1 \leq 2$.

2. **Degree of Scalar Multiplication:** $\deg(\lambda P) = \deg(P)$.
3. **Equality of Polynomials:** If $P = Q$, then $\deg(P) = \deg(Q)$.
4. **Integral Domain:** The ring $K[X]$ is an integral domain. This means for $P, Q \in K[X]$, if $PQ = 0$ then $P = 0$ or $Q = 0$.
5. **Product of Polynomials:** For non-zero $P, Q \in K[X]$, $\deg(PQ) = \deg(P) + \deg(Q)$. If $P = 0$ or $Q = 0$, then $PQ = 0$ and $\deg(P) + \deg(Q) = -\infty$.
6. **Units in $K[X]$:** The units are the non-zero constant polynomials. $(K[X])^\times = K \setminus \{0_K\}$.

- **Proof**

Let $P, Q \in K[X]$ such that $PQ = 1$. Then $P \neq 0$ and $Q \neq 0$. We have

$\deg(P) + \deg(Q) = \deg(PQ) = \deg(1) = 0$. Since $\deg(P), \deg(Q) \in \mathbb{N}_0$, the only solution is $\deg(P) = 0$ and $\deg(Q) = 0$. Thus, P and Q are non-zero constants. $\square$

Valuation of a Polynomial

Let $P = \sum_{k \geq 0} a_k X^k \in K[X]$. The **valuation** of P , denoted by $\text{val}(P)$, is defined as:

$$\text{val}(P) = \begin{cases} \min\{n \in \mathbb{N}_0 \mid a_n \neq 0\} & \text{if } P \neq 0 \\ +\infty & \text{if } P = 0 \end{cases}$$

For two polynomials P and Q , we have: $\text{val}(PQ) = \text{val}(P) + \text{val}(Q)$. In particular, $\text{val}(\lambda P) = \text{val}(P)$ for $\lambda \in K \setminus \{0_K\}$.

1.4 Composition and Derivatives

Composition of Two Polynomials

Let $P = \sum_{k \geq 0} a_k X^k$ and $Q \in K[X]$. The polynomial $\sum_{k \geq 0} a_k Q^k$ is the composition of Q by P and is denoted by $P \circ Q$.

Example

If $P = 1 + X^2$ and $Q = X + X^3$, then $P \circ Q = P(Q) = 1 + Q^2 = 1 + (X + X^3)^2$.

Properties of Composition

Let $P, Q, R \in K[X]$ and $\lambda, \mu \in K$. Then:

1. $(\lambda P + \mu Q) \circ R = \lambda(P \circ R) + \mu(Q \circ R)$
2. $(PQ) \circ R = (P \circ R)(Q \circ R)$
3. If $\deg(Q) \geq 1$, then $\deg(P \circ Q) = \deg(P) \cdot \deg(Q)$.

Exercise: Show that $\log(x)$ is Not a Polynomial

For the sake of contradiction, assume $\log(x)$ is a polynomial function associated with a non-constant polynomial $P(X) \in \mathbb{R}[X]$. The functional identity $\log(x^2) = 2\log(x)$ for $x > 0$ implies a polynomial identity $P(X^2) = 2P(X)$. Comparing degrees: $\deg(P(X^2)) = \deg(P) \cdot \deg(X^2) = 2\deg(P)$. $\deg(2P(X)) = \deg(P)$. So, $2\deg(P) =$

$\deg(P)$, which implies $\deg(P) = 0$. This means $P(X)$ must be a constant, say $P(X) = c$. This contradicts that the function is not constant (e.g., $\log(1) = 0 \neq \log(e) = 1$).

Derivative of a Polynomial

Let $P = \sum_{k \geq 0} a_k X^k \in K[X]$. The **derivative** of P , denoted by P' , is:

$$P' = \sum_{k \geq 1} k a_k X^{k-1}$$

For higher derivatives: $P^{(0)} = P$, and $P^{(n+1)} = (P^{(n)})'$ for all $n \geq 0$.

Example

If $P = 2X^3 + X^2 - 3X + 1$:

- $P' = 6X^2 + 2X - 3$
- $P'' = 12X + 2$
- $P^{(3)} = 12$
- $P^{(n)} = 0$ for all $n \geq 4$

Properties of Derivatives

Let $P, Q \in K[X]$ and $n \in \mathbb{N}_0$:

1. If $\deg(P) \geq n$, then $\deg(P^{(n)}) = \deg(P) - n$.
2. If $\deg(P) < n$, then $P^{(n)} = 0$, so $\deg(P^{(n)}) = -\infty$.
3. $(P + Q)^{(n)} = P^{(n)} + Q^{(n)}$
4. $(PQ)^{(n)} = \sum_{k=0}^n \binom{n}{k} P^{(k)} Q^{(n-k)}$ (Leibniz formula)
5. $(P \circ Q)' = Q' \cdot (P' \circ Q)$ (Chain rule)

Taylor's Formula for Polynomials

Let $P \in K[X]$ of degree $n \geq 0$ and let $\alpha \in K$. Then:

$$P(X) = \sum_{k=0}^n \frac{P^{(k)}(\alpha)}{k!} (X - \alpha)^k$$

Remark

If $\alpha = 0$, Taylor's formula gives back the original polynomial definition:

$$P(X) = \sum_{k=0}^n \frac{P^{(k)}(0)}{k!} X^k$$

since $P^{(k)}(0) = k!a_k$, where a_k is the coefficient of X^k .

1.5 Polynomial Functions and Evaluation**Value of a Polynomial at λ**

Let $P = \sum_{k \geq 0} a_k X^k \in K[X]$ and let $\lambda \in K$. The value of P at λ is:

$$P(\lambda) = \sum_{k \geq 0} a_k \lambda^k \in K$$

In $\mathbb{Z}_5$

Let $K = \mathbb{Z}_5$ and $P = 1 + 3X + 2X^3$.

$$P(2) = 1 + 3 \cdot 2 + 2 \cdot 2^3 = 1 + 6 + 16 \equiv 1 + 1 + 1 = 3 \pmod{5}$$

Polynomial Function

Let $P \in K[X]$. The function $\tilde{P} : K \rightarrow K$, $x \mapsto P(x)$ is called the **polynomial function** associated with the polynomial P .

Example

Let $P \in \mathbb{R}[X]$ be defined by $P = \sqrt{2} - X + \frac{2}{3}X^2$. The associated polynomial function is:

$$\tilde{P} : \mathbb{R} \rightarrow \mathbb{R}, \quad x \mapsto \sqrt{2} - x + \frac{2}{3}x^2$$

2. Divisibility and Division in $K[X]$

2.1 Divisibility

Divisibility Factor Divisor

Let $A, B \in K[X]$ with $B \neq 0$. We say that B **divides** A (or is a divisor/factor of A), and we write $B \mid A$, if $A = PB$ for some $P \in K[X]$.

Example

1. $X + 1 \mid X^2 + 3X + 2$ in $\mathbb{R}[X]$ because $X^2 + 3X + 2 = (X + 1)(X + 2)$.

2. $2X^2 + X + 1 \mid 2X^3 + 2X + 1$ in $\mathbb{Z}_3[X]$ because in $\mathbb{Z}_3[X]$:

$$(2X^2 + X + 1)(X + 1) = 2X^3 + 2X^2 + X^2 + X + X + 1 = 2X^3 + 3X^2 + 2X + 1 = 2X^3 + 2X + 1$$

Properties of Divisibility

Let $A, B, C, D \in K[X]$:

1. If $A \mid B$ and $B \mid A$, then $A = \lambda B$ for some $\lambda \in K^\times$.
2. If $D \mid A$ and $D \mid B$, then $D \mid (\lambda A + \mu B)$ for all $\lambda, \mu \in K[X]$.
3. If $A \mid B$ and $C \mid D$, then $AC \mid BD$.

2.2 Polynomial Division Theorem

Euclidean Division

Let $A, B \in K[X]$ with $B \neq 0$. Then, there exist unique polynomials $Q, R \in K[X]$ such that:

$$A = B \cdot Q + R$$

where $\deg(R) < \deg(B)$. Q is called the quotient and R is the remainder.

Division in $\mathbb{R}[X]$

Let's divide $A(X) = X^3 + 2X + 1$ by $B(X) = X^2 - 1$. Using long division:

$$\begin{array}{r} X \\ X^2 - 1 \overline{) X^3 + 0X^2 + 2X + 1} \\ \underline{-(X^3 \quad - X)} \\ 0X^3 + 3X + 1 \end{array}$$

Thus, the quotient is $Q(X) = X$ and the remainder is $R(X) = 3X + 1$.

$$X^3 + 2X + 1 = (X^2 - 1) \cdot X + (3X + 1)$$

Division in $\mathbb{Z}_5[X]$

Let's divide $A(X) = 2X^3 + 4X + 1$ by $B(X) = 2X + 2$ in $\mathbb{Z}_5[X]$.

$$\begin{array}{r}
 X^2 + 4X + 3 \\
 2X + 2 \overline{) 2X^3 + 0X^2 + 4X + 1} \\
 \underline{-(2X^3 + 2X^2)} \\
 -2X^2 + 4X \equiv 3X^2 + 4X \\
 \underline{-(3X^2 + 3X)} \\
 X + 1 \\
 \underline{-(X + 1)} \\
 0
 \end{array}$$

Thus, $2X^3 + 4X + 1 = (2X + 2)(X^2 + 4X + 3)$ in $\mathbb{Z}_5[X]$. The remainder is 0.

2.3 Division by Increasing Power Order

Division by Increasing Power Order

Let $A, B \in K[X]$ such that $B(0) \neq 0$. For any $n \in \mathbb{N}$, there exist unique polynomials Q and R in $K[X]$ such that:

$$A = BQ + X^{n+1}R$$

with $\deg(Q) \leq n$. Q is the quotient and R is the remainder. The integer n is the order of division.

Example

Let $A(X) = 1 + X$ and $B(X) = 1 - X$. Find the quotient to order $n = 2$.

$1 + X$	$1 - X$
$-(1 - X)$	$1 + 2X + 2X^2$
$2X$	
$-(2X - 2X^2)$	
$2X^2$	
$-(2X^2 - 2X^3)$	
$2X^3$	

So, $1 + X = (1 - X)(1 + 2X + 2X^2) + 2X^3$. Here, $Q(X) = 1 + 2X + 2X^2$ and the remainder term is $X^{2+1}R(X) = 2X^3$, so $R(X) = 2$.

3. Roots and Factorization

3.1 Roots of Polynomials

Root of a Polynomial

Let $P \in K[X]$ and $\alpha \in K$. α is said to be a **root** (or a **zero**) of P in K if $P(\alpha) = 0$.

Examples

1. 2 is a zero of the polynomial $-X^3 + 2X + 4$ in $\mathbb{R}$ because: $-2^3 + 2(2) + 4 = -8 + 4 + 4 = 0$.
2. i is a root of the polynomial $X^2 + 1$ in $\mathbb{C}$ because: $i^2 + 1 = -1 + 1 = 0$.
3. The roots of $X^n - 1$ in $\mathbb{C}$ are $e^{2\pi i k/n}$ for $k = 0, \dots, n-1$.
4. The roots of $X^3 + X^2 + X + 1 = \frac{X^4-1}{X-1}$ are the 4th roots of unity, except 1. They are $i, -1, -i$.

Notable Values

- Powers of -1 cycle through $\{1, -1\}$.
- Powers of $j = e^{2\pi i/3}$ cycle through $\{1, j, j^2\}$.
- Powers of i cycle through $\{1, i, -1, -i\}$.

Remainder Theorem

Let $P \in K[X]$ and $\alpha \in K$. The remainder of the Euclidean division of $P(X)$ by $X - \alpha$ is the constant $P(\alpha)$.

• **Proof**

By the division algorithm, $P(X) = Q(X)(X - \alpha) + R(X)$ with $\deg(R) < \deg(X - \alpha) = 1$. Thus, $R(X)$ must be a constant, say $R(X) = c \in K$. Evaluating at $X = \alpha$ gives $P(\alpha) = Q(\alpha)(\alpha - \alpha) + c = c$. Thus, the remainder is $P(\alpha)$. $\square$

Corollary

1. $X - \alpha \mid P(X) \iff P(\alpha) = 0$.
2. If $A \mid B$, then all roots of A are also roots of B .

Multiplicity of a Zero

Let $P \in K[X]$ be a nonzero polynomial, and let $\alpha \in K$ be a zero of P . The **multiplicity** of α is the largest positive integer r such that $(X - \alpha)^r$ divides P .

- A **simple zero** has multiplicity 1.
- A **double zero** has multiplicity 2.
- A **triple zero** has multiplicity 3.

Equivalent Conditions for Multiplicity

Let $\alpha \in K$ and $P(X) \in K[X]$ be non-zero, and $r \in \mathbb{Z}^+$. The following statements are equivalent:

1. α is a zero of P with multiplicity r .
2. There exists $Q(X) \in K[X]$ such that $P(X) = (X - \alpha)^r Q(X)$ and $Q(\alpha) \neq 0$.
3. $P(\alpha) = P'(\alpha) = \dots = P^{(r-1)}(\alpha) = 0$, and $P^{(r)}(\alpha) \neq 0$.

Checking Multiplicity

Consider the polynomial:

$$P(x) = 5x^5 - 20x^4 + 5x^3 + 50x^2 - 20x - 40$$

Dividing by $(x - 2)$ using polynomial long division:

	$5x^4$	$-10x^3$	$-15x^2$	$20x$	20	
$x - 2$	$5x^5$	$-20x^4$	$+5x^3$	$+50x^2$	$-20x$	-40
	$-5x^5$	$+10x^4$				
		$-10x^4$	$+5x^3$			
		$10x^4$	$-20x^3$			
			$-15x^3$	$+50x^2$		
			$15x^3$	$-30x^2$		
				$20x^2$	$-20x$	
				$-20x^2$	$+40x$	
					$20x$	-40
					$-20x$	$+40$
						0

Thus, the division shows that:

$$P(x) = (x - 2)(5x^4 - 10x^3 - 15x^2 + 20x + 20)$$

Conjugate Roots

Let $P(X) \in \mathbb{R}[X]$. If $\alpha \in \mathbb{C}$ is a zero of P with multiplicity r , then its complex conjugate $\bar{\alpha}$ is also a zero of P with the same multiplicity r .

Polynomial Zero Structure

Let $P(X) \in K[X]$ of degree $n \geq 0$. Let $\alpha_1, \dots, \alpha_p \in K$ be distinct zeros of P with multiplicities $r_1, \dots, r_p$. Then there exists $Q(X) \in K[X]$ such that:

$$P(X) = (X - \alpha_1)^{r_1}(X - \alpha_2)^{r_2} \dots (X - \alpha_p)^{r_p}Q(X)$$

where $Q(\alpha_i) \neq 0$ for all $i = 1, \dots, p$. Furthermore, $\sum r_i \leq \deg(P)$.

Corollary

A non-zero polynomial of degree n has at most n roots in K , counted with multiplicity. A polynomial that has more zeros than its degree must be the zero polynomial.

3.2 Factorization in $\mathbb{C}[X]$ and $\mathbb{R}[X]$

Non-polynomial functions

The function $\sin(x)$ is not a polynomial function because it has infinitely many roots ($k\pi$ for $k \in \mathbb{Z}$), but a non-zero polynomial can only have a finite number of roots. All non-constant periodic functions are not polynomials.

Irreducible and Reducible Polynomials

- A non-constant polynomial $P \in K[X]$ is **irreducible** over K if it cannot be factored into the product of two non-constant polynomials with coefficients in K .
- If P is not irreducible, it is **reducible**.

Examples

1. Every polynomial of degree 1 is irreducible.
2. The trinomial $aX^2 + bX + c \in \mathbb{R}[X]$ is irreducible over $\mathbb{R}$ if $b^2 - 4ac < 0$.
3. Any polynomial $aX^2 + bX + c \in \mathbb{R}[X]$ is reducible over $\mathbb{C}$.

Fundamental Theorem of Algebra (D'Alembert's Theorem)

Every non-constant polynomial in one variable with complex coefficients has at least one complex root.

Corollaries

- The irreducible polynomials in $\mathbb{C}[X]$ are precisely the polynomials of degree 1.
- A polynomial of degree $n \geq 1$ has exactly n roots in $\mathbb{C}$, counted with multiplicity.

Factorization over $\mathbb{C}$

Let $P(X) \in \mathbb{C}[X]$ be a polynomial of degree $n \geq 1$. Then $P(X)$ can be factored uniquely (up to ordering) as:

$$P(X) = C(X - \alpha_1)^{r_1}(X - \alpha_2)^{r_2} \cdots (X - \alpha_k)^{r_k}$$

where C is the leading coefficient, α_i are the distinct roots, and $\sum r_i = n$.

Examples

1. Factorize $2X^3 - 2$:

$$2X^3 - 2 = 2(X^3 - 1) = 2(X - 1)(X - j)(X - j^2)$$

where $j = e^{2\pi i/3}$ is a primitive cube root of unity.

2. Factorize $X^4 + 1$: The roots are solutions to $X^4 = -1 = e^{i(\pi+2k\pi)}$. They are $e^{i\pi/4}, e^{i3\pi/4}, e^{i5\pi/4}, e^{i7\pi/4}$.

$$X^4 + 1 = (X - e^{i\pi/4})(X - e^{i3\pi/4})(X - e^{i5\pi/4})(X - e^{i7\pi/4})$$

Irreducible Polynomials in $\mathbb{R}[X]$

A polynomial $P(X) \in \mathbb{R}[X]$ is irreducible over $\mathbb{R}$ if and only if:

1. $P(X)$ has degree 1, or
2. $P(X)$ has degree 2 with a negative discriminant ($b^2 - 4ac < 0$).

Factorize $X^4 + 1$ in $\mathbb{R}[X]$

We pair the complex conjugate roots from the factorization in $\mathbb{C}[X]$. Note that $\overline{e^{i\pi/4}} = e^{-i\pi/4} = e^{i7\pi/4}$ and $\overline{e^{i3\pi/4}} = e^{-i3\pi/4} = e^{i5\pi/4}$.

$$\begin{aligned} (X - e^{i\pi/4})(X - e^{-i\pi/4}) &= X^2 - (e^{i\pi/4} + e^{-i\pi/4})X + 1 = X^2 - 2\cos(\pi/4)X + 1 \\ &= X^2 - \sqrt{2}X + 1 \end{aligned}$$

$$\begin{aligned} (X - e^{i3\pi/4})(X - e^{-i3\pi/4}) &= X^2 - (e^{i3\pi/4} + e^{-i3\pi/4})X + 1 = X^2 - 2\cos(3\pi/4)X + 1 \\ &= X^2 + \sqrt{2}X + 1 \end{aligned}$$

Thus, $X^4 + 1 = (X^2 - \sqrt{2}X + 1)(X^2 + \sqrt{2}X + 1)$. The discriminants of $X^2 - \sqrt{2}X + 1$ and $X^2 + \sqrt{2}X + 1$ are both $-2 < 0$, so they are irreducible over $\mathbb{R}$.

Factorize $X^3 + 27$ in $\mathbb{R}[X]$

Using the sum of cubes formula $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$:

$$X^3 + 27 = X^3 + 3^3 = (X + 3)(X^2 - 3X + 9)$$

The factor $X + 3$ is irreducible (degree 1). The factor $X^2 - 3X + 9$ has discriminant $\Delta = (-3)^2 - 4(1)(9) = -27 < 0$, so it is also irreducible over $\mathbb{R}[X]$.

Factorize $X^8 + 2X^6 + 3X^4 + 2X^2 + 1$ in $\mathbb{R}[X]$

This is a reciprocal polynomial. We can see this factors as $(X^4 + X^2 + 1)^2$. Factoring the inner part:

$$X^4 + X^2 + 1 = (X^4 + 2X^2 + 1) - X^2 = (X^2 + 1)^2 - X^2 = (X^2 - X + 1)(X^2 + X + 1)$$

Both quadratic factors have discriminant $-3 < 0$ and are irreducible over $\mathbb{R}$. The final factorization is:

$$(X^2 - X + 1)^2(X^2 + X + 1)^2$$

4. GCD and LCM of Polynomials

4.1 The Euclidean Algorithm

Existence of GCD

For any two polynomials $A, B \in K[X]$, not both zero, their **greatest common divisor (GCD)** exists. It is the unique monic polynomial D that satisfies:

1. D is a common divisor of A and B .
2. Any other common divisor D' of A and B also divides D .

Euclidean Algorithm for GCD

Let A, B be two polynomials with $B \neq 0$. If R is the remainder of the Euclidean division of A by B , then:

$$\gcd(A, B) = \gcd(B, R)$$

By repeating this process until a zero remainder is found, the last non-zero remainder (made monic) is the GCD.

GCD Calculation

Find $\gcd(A, B)$ for $A = X^5 - 2X^4 + X^3 - X^2 + 2X - 1$ and $B = X^3 - X^2 + 2X - 2$.

1. Divide A by B : $A = B \cdot (X^2 - X - 1) + (X^2 + 4X - 5)$. Let $R_1 = X^2 + 4X - 5$.
2. Divide B by R_1 : $B = R_1 \cdot (X - 5) + (27X - 27)$. Let $R_2 = 27X - 27$.
3. Divide R_1 by R_2 : Since $R_2 = 27(X - 1)$ and $R_1 = (X - 1)(X + 5)$, it is clear

that $R_2 \mid R_1$. The next remainder is 0.

The last non-zero remainder is $R_2 = 27(X - 1)$. The monic version is $X - 1$. Thus, $\gcd(A, B) = X - 1$.

4.2 Bézout's Theorem and the Extended Euclidean Algorithm

Bézout's Theorem

For any two non-zero polynomials $A, B \in K[X]$, there exist polynomials $U, V \in K[X]$ such that:

$$UA + VB = \gcd(A, B)$$

The polynomials U and V can be found by reversing the steps of the Euclidean algorithm.

Finding Bézout Coefficients

From the previous example, we have:

$$\begin{aligned} R_1 &= A - (X^2 - X - 1)B \\ 27(X - 1) &= B - (X - 5)R_1 \end{aligned}$$

Substitute R_1 into the second equation:

$$\begin{aligned} 27(X - 1) &= B - (X - 5)[A - (X^2 - X - 1)B] \\ &= B - (X - 5)A + (X - 5)(X^2 - X - 1)B \\ &= (-X + 5)A + [1 + (X - 5)(X^2 - X - 1)]B \\ &= (-X + 5)A + (1 + X^3 - X^2 - X - 5X^2 + 5X + 5)B \\ &= (-X + 5)A + (X^3 - 6X^2 + 4X + 6)B \end{aligned}$$

Dividing by 27 gives the coefficients for the monic GCD: $U = \frac{1}{27}(-X + 5)$ and $V = \frac{1}{27}(X^3 - 6X^2 + 4X + 6)$.

Coprime Polynomials

Two polynomials $A, B \in K[X]$ are **coprime** (or relatively prime) if $\gcd(A, B) = 1$.

Gauss's Lemma for Polynomials

Let $A, B, C \in K[X]$. If $A \mid BC$ and $\gcd(A, B) = 1$, then $A \mid C$.

4.3 GCD, LCM, and Factorization**GCD from Roots**

Given the factorization of $A, B \in \mathbb{C}[X]$, the GCD is the product of the common factors, each raised to the minimum of its multiplicities in A and B .

Example

Let:

- $A = (X - 3)^3(X - 1)^5(X + 2)^3(X^2 + 1)^2$
- $B = (X - 5)^3(X - 1)^2(X + 2)^7(X^2 + 1)(X^2 + X + 1)$

The common prime factors are $(X - 1)$, $(X + 2)$, and $(X^2 + 1)$.

- Power of $(X - 1)$: $\min(5, 2) = 2$
- Power of $(X + 2)$: $\min(3, 7) = 3$
- Power of $(X^2 + 1)$: $\min(2, 1) = 1$

So, $\gcd(A, B) = (X - 1)^2(X + 2)^3(X^2 + 1)$.

LCM and other Properties

For monic polynomials $A, B \in K[X]$:

$$\gcd(A, B) \cdot \text{lcm}(A, B) = A \cdot B$$

For any non-zero constants $x, y \in K$:

$$\gcd(A, B) = \text{monic}(\gcd(xA, yB))$$

This course material was prepared by Prof Rihane and transcribed by students @sidali Boulil. You may encounter minor typos or errors due to transcription. We appreciate your understanding and hope you find these notes useful.